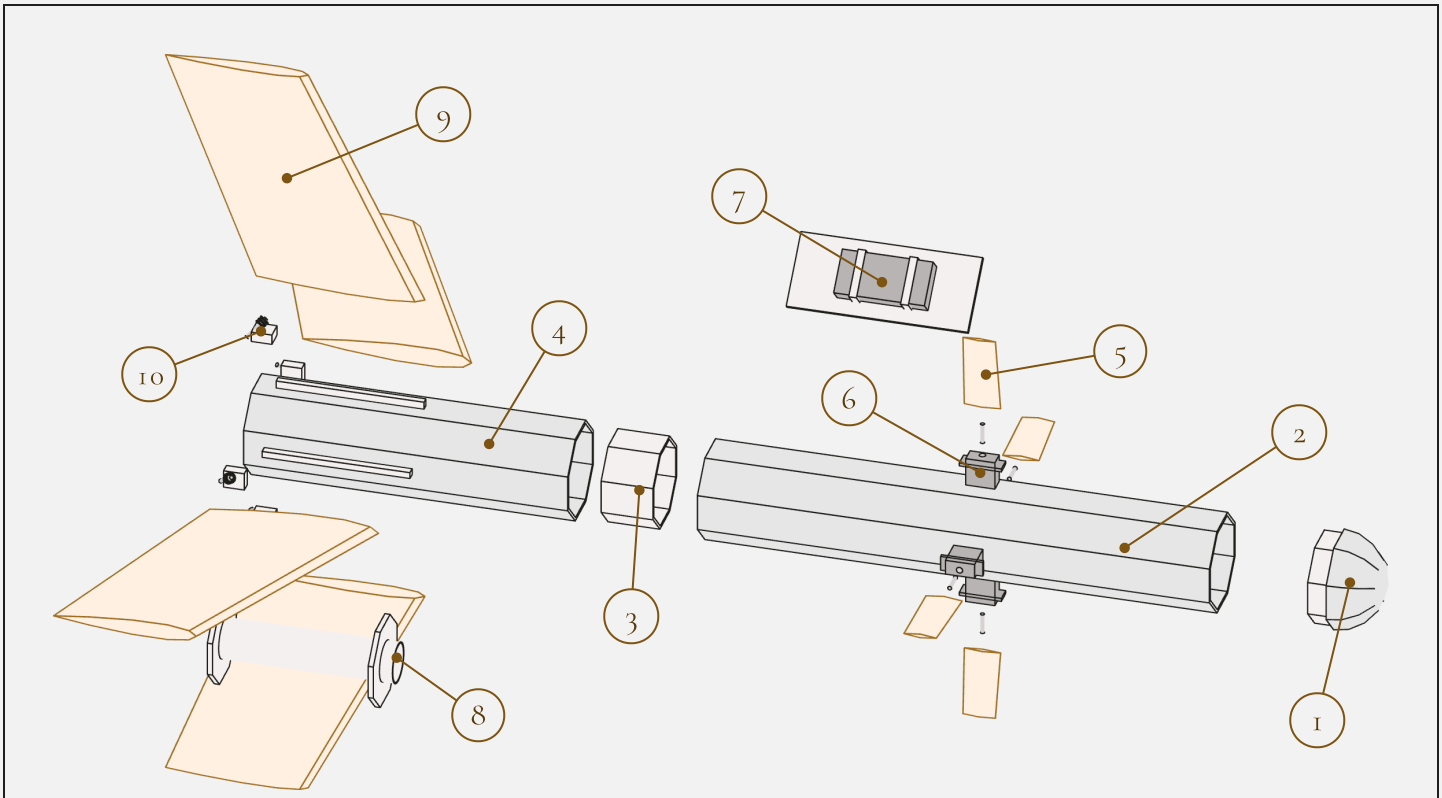
FLIGHT-MANUAL DISCIPLINE × GROUNDED AI × OPEN-SOURCE ROCKET

Project 33 *Issue B*



FO-1	EXPLODED ASSEMBLY · FOLDING-FIN ROCKET	UNITS MM	VISUALIZATION, NOT EVIDENCE

1 Nose cone, faceted
2 Forward body tube
3 Coupler and bulkhead
4 Aft body tube
5 Canards ×4
6 SG90 canard servos ×4
7 Avionics sled: ESP32, MPU6050
8 Motor mount, centering rings
9 Folding fins ×4, NACA 0012
10 Hinges, torsion springs

LENGTH	DIAMETER	STABILITY	EST. COST
68.3 cm	7.62 cm	1.03 cal	$81

Build a model. Make a prediction. Measure the difference. Keep the record honest.

OPEN SOURCE · MIT github.com/topherchris420/33

Code, CAD, firmware, models and the full evidence record.

topherchris420.github.io/33

Three sources, *one record*

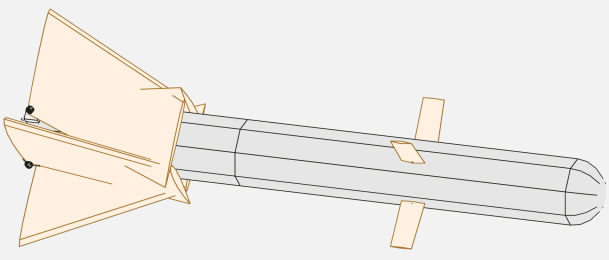
№	SOURCE	BORROWED TECHNOLOGY	IN PROJECT 33, ISSUE B	BUILDS ON
01	F-117A	Prestart built-in test	Numbered power-on self-test on both ESP32 boards, logged to the session	Preflight audit, USB bench monitor
02	F-117A	Dual-channel autopilot, channel-2 reversion	A declared fallback state for every sensor or link loss	Degraded-mode matrix
03	F-117A	Annunciator panel, master caution	Launcher master-caution LED and buzzer, driven by advisory severity	Ground-status board
04	F-117A	INS with dead-reckoning and AHRS fallback	Attitude-source ladder with a validity flag in every packet	MPU6050 telemetry
05	F-117A	PAARS automatic recovery	Pressure-based apogee detection, reviewed offline before any actuation	BMP180 altimeter review
06	F-117A	Effective pages, OFP-45 configuration control	Issue-stamped manual: each page carries the digests of the source it describes	Bench manual SHA-256 index
07	REACTOR EXPERT 3B	Grounded tool layer	Numeric questions route to the committed models; the language model only explains their outputs	Evidence query, analytical models
08	REACTOR EXPERT 3B	Fail-closed routing	Unknown or malformed requests fall back to the deterministic path, never to a guess	Evidence check, safety register
09	REACTOR EXPERT 3B	Response contract	Every answer states boundaries, governing model, analysis, safety and limitations	Claim passports

AUTHORITY ORDER · ISSUE B EVIDENCE ASSISTANT

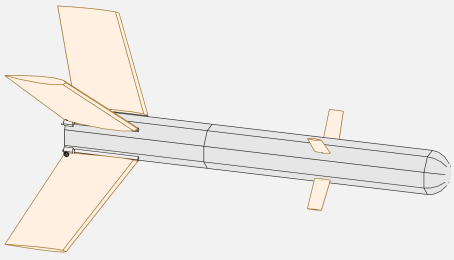
Safety gates › Input validation › Verified models › Claim ledger › Language model | *A named human accepts*

Adapted from the Civil Reactor Expert 3B: code computes, the model explains, and no answer outranks a gate. Issue B borrows the architecture, not the adapter's weights, which are a 5-step validation run.

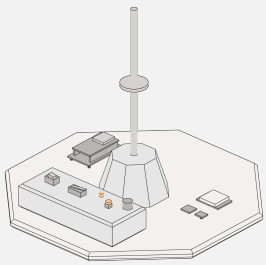
FO-2 · CONFIGURATIONS



Stowed: each fin turns 120° on a skewed hinge



Deployed: 130/110 × 180 mm fins



Launcher platform

Issue B items are proposals, not built or measured. Aircraft procedures and limits do not transfer. Drawings are generated from the repository's model.html geometry.

Sources: TO 1F-117A-1A, OCIP supplement (1992) · github.com/topherchris420/33 (MIT) · huggingface.co/ciaochris/Vers3Dynamics-Civil-Reactor-Expert-3B